

⚙️ M002 Week 7 — Pistons

Topic: Pistons and Mazes · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

A piston pushes or pulls a block when it gets a redstone signal. A lever sends that signal. Flip the lever, the piston opens a path across a gap, and the agent crosses while the path is open.

1 · Predict 🧠

Read the steps. Write what happens.

```
move forward
interact ahead
move forward
```

The agent reaches a lever and flips it. A piston opens the path. Can the agent cross now?


```
if gap ahead:
    interact ahead
otherwise:
    move forward
```

The agent is on solid ground, no gap. What does it do?

2 · Trace 🦶

```
keep doing forever:
  if gap ahead:
    interact ahead
    move forward
  otherwise:
    move forward
```

Say in plain English what the agent does at a gap, and what it does on solid ground.

3 · Fill the Code Blank

Now in real Python. The agent flips the lever only when it sees a gap ahead. Fill the blank.

```
if not agent.detect(BLOCK, FORWARD):  
    agent.____(...)  
    agent.move(FORWARD, 1)  
else:  
    agent.move(FORWARD, 1)
```

Word bank: interact · turn · detect

Write the missing word:

4 · Spot One Bug 🐛

The agent should power the piston **first**, then cross the gap it opens. This code crosses first.

```
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
agent.interact(...)
```

Write the fixed code (flip the lever before the gap):

One sentence: why was it wrong?

5 · Build It + Plan Your Video 📷 🎥

Open your homework world. The maze has a gap and a piston that opens it. Walk the agent to the lever, flip it, cross while the path is open, and reach the goal.

Send a photo or video of the agent at the end. Then finish these lines.

I powered the piston by ...

Once the path opened, the agent ...

The hardest part was ...

Take a phone video while the agent runs. Talk like you are teaching a friend. Use these words: **piston**, **redstone**, **signal**, **gap**.

1. Show the gap and the piston.
2. Run your code and show the agent crossing.
3. Say which line opens the path.

Write what you will say:

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.